THREE NODES, ONE CORRIDOR – NIAGARA SOVEREIGN AI POWER BELT

Delivering 1 GW+ of Sovereign AI Compute on Crown Land and Clean Power.

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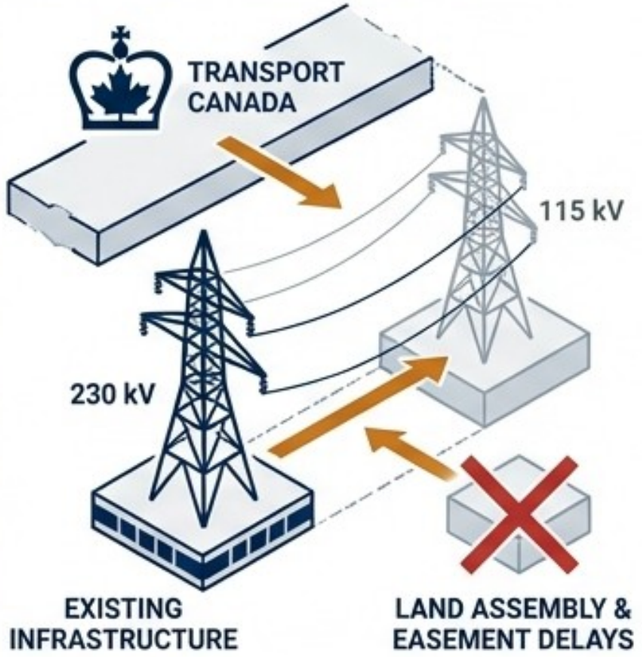
EXECUTIVE OVERVIEW

THE CONCEPT

A phased 1 GW+ AI compute corridor linking Allanburg, Welland, and Port Colborne via the federal Welland Canal infrastructure spine.

THE ADVANTAGE

Utilizes existing Crown/Transport Canada land and high-voltage transmission (230 kV/115 kV) to bypass typical land-assembly and easement delays.



THE GOAL

Answer the federal 'Sovereign AI Compute' mandate ($2B+ strategy) by creating a secure, domestic infrastructure belt powered by Niagara's clean energy.

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THE PROBLEM & THE OPPORTUNITY

THE OPPORTUNITY

The Mandate: Canada has committed ~$2B+ to sovereign AI, requiring data residency, domestic control, and clean power.

The Bottleneck: Private land assembly is fragmented; grid connections are congested; foreign hyperscalers dominate non-sovereign clouds.

The Solution: Activate dormant industrial Crown lands along the canal to create a "public-led" corridor for secure, high-density compute and energy storage.

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CONCEPT MAP – ‘THREE NODES, ONE CORRIDOR’

Utilizing Transport Canada/Seaway lands for a unified fiber/power Right-of-Way.

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ALLANBURG STRATEGIC ALIGNMENT (PHASE 1)

Primary Asset: Direct line-of-sight to Allanburg TS (major 230 kV hub), minimizing transmission interconnection costs and time.

Land Strategy: “Rings of Control” — Core industrial build pad surrounded by federal/public buffer lands.

Sovereignty: Anchored by Crown land, ensuring long-term Canadian control over critical infrastructure easements.

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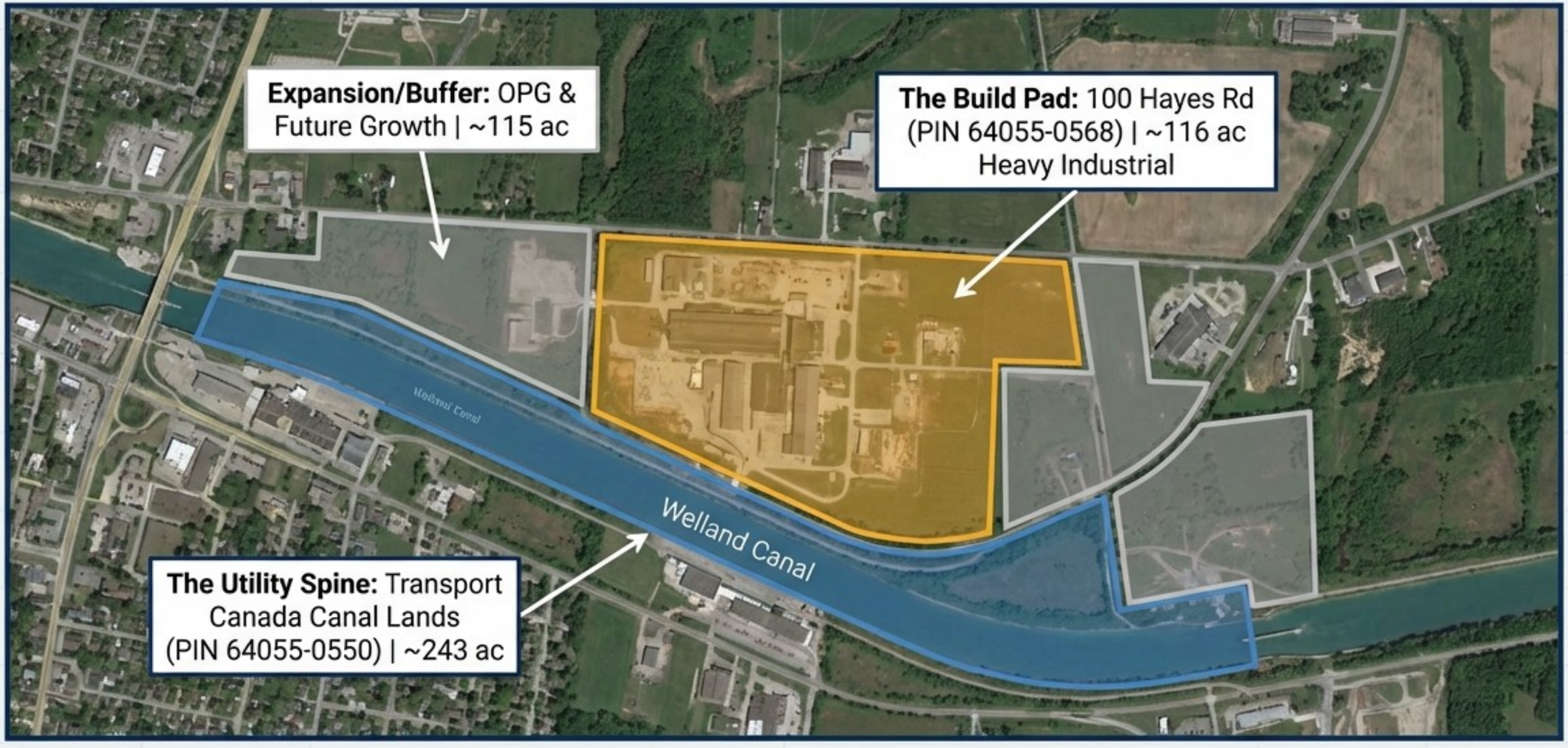
THE ALLANBURG LAND STACK



Total Scope: >470 acres of controlled or influenced land within the immediate power zone.

WHY ALLANBURG IS THE IDEAL ‘START NODE’

SPEED TO MARKET

Hayes Rd is an existing industrial site, not greenfield farmland; Zoning is already HI (Heavy Industrial).

POWER ACCESS

Distance to Allanburg TS is <1 km (via Canal easement); supports immediate 100–200 MW injection.

RESILIENCE

Space for on-site BESS (Battery Energy Storage) to stabilize local grid and support high-density AI loads.

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PHASE 2 – WELLAND AI + BESS NODE

- **~79 ac Heavy Industrial** site (GEC-72/L1 zoning).
- Infrastructure: High assessment value (**$64M+**) indicates **heavy servicing** (water, power, road) is already in place.
- Proximity: Situated minutes from **Welland TS**; ideal for mid-stage expansion (**+300 MW**).

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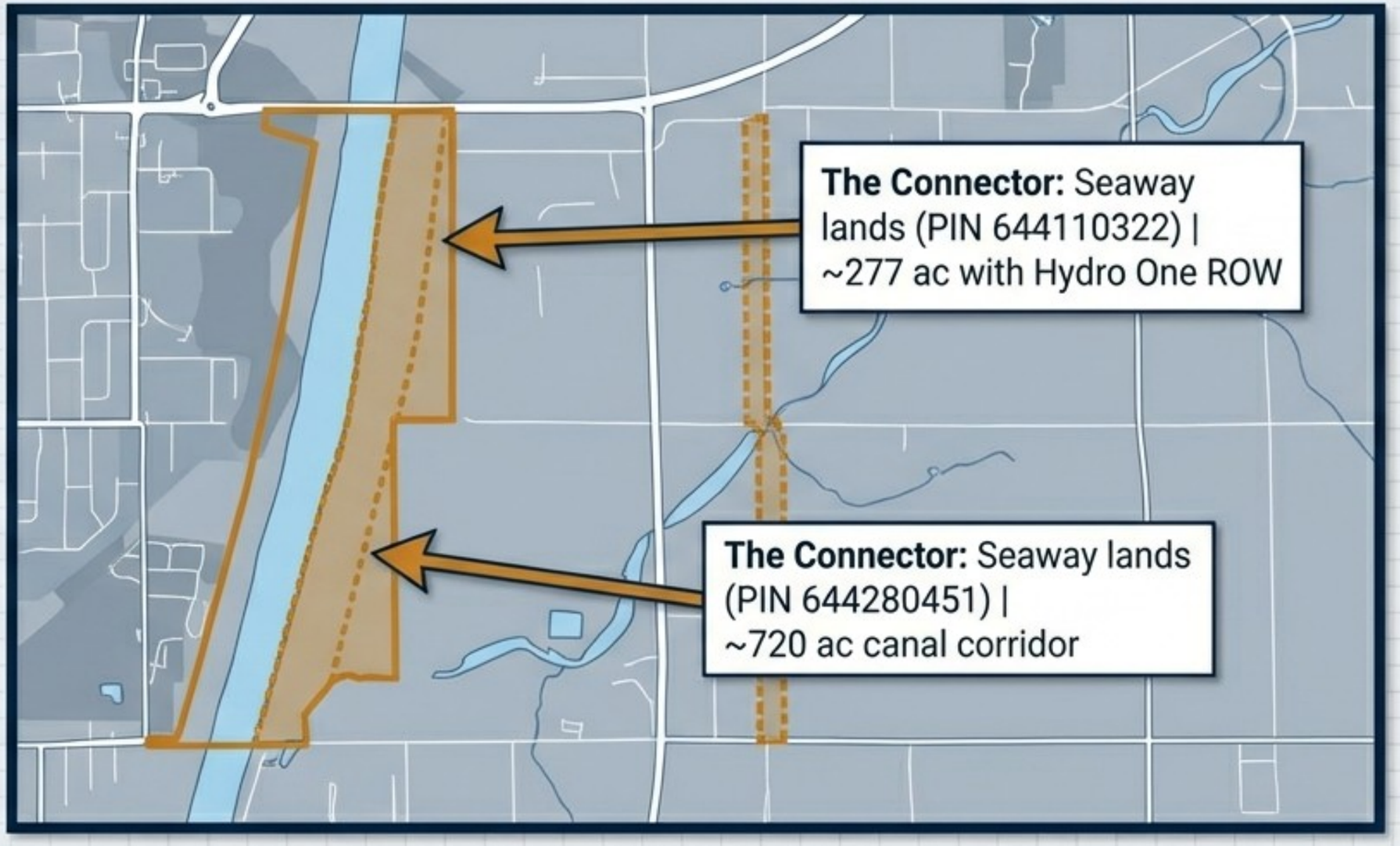
THE SEAWAY & HYDRO ONE CORRIDOR



Strategy: Use federal Seaway lands for '**sovereign easements**' to link the Buchner campus to the grid without private land expropriation.

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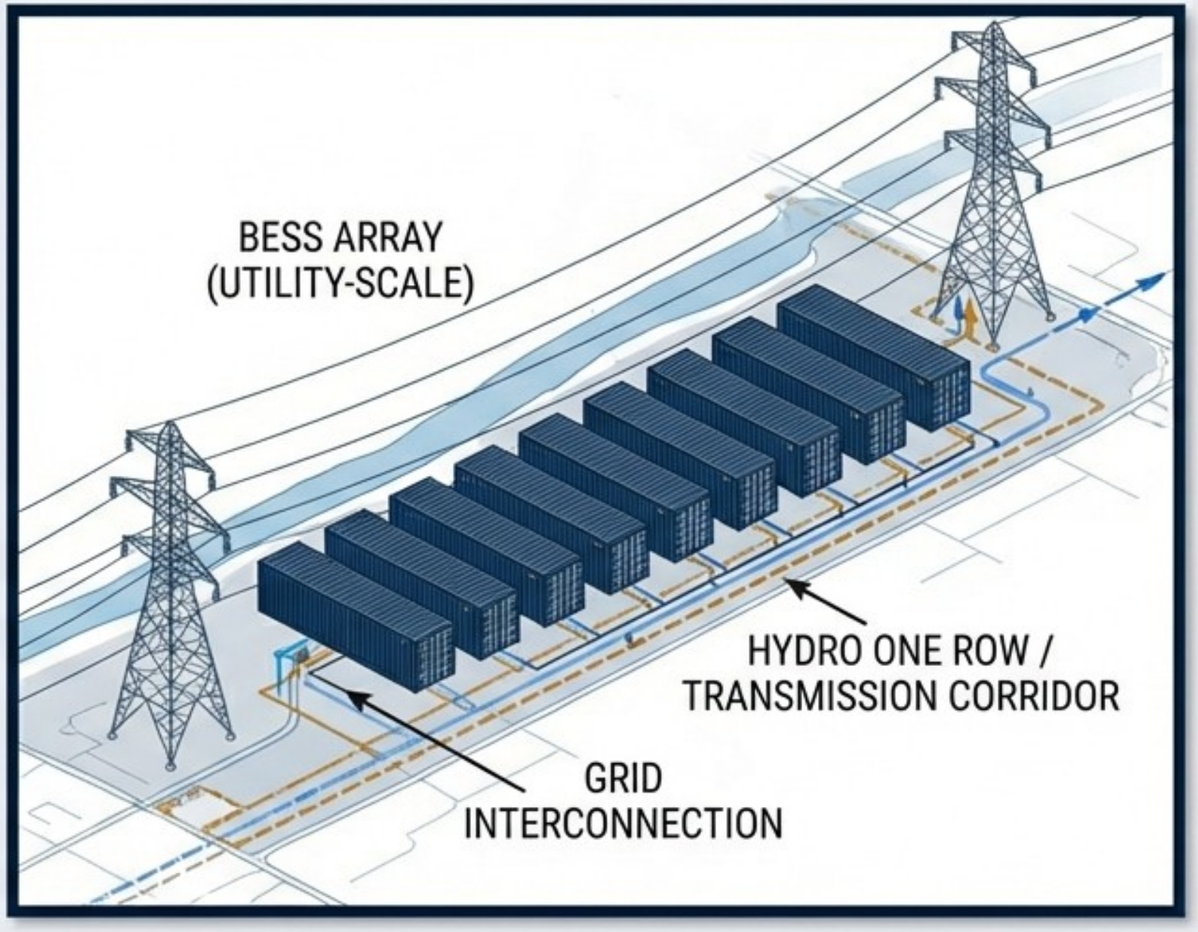
INDIGENOUS PARTNERSHIP & STORAGE



- **Opportunity:** The Seaway/Hydro corridor is ideal for an **Indigenous-equity transmission and BESS partnership** (modelled on Niagara Reinforcement Line).
- **Grid Role:** Utility-scale **BESS deployed along the Hydro One ROW** provides frequency regulation and backup for the AI cluster.
- **Economic Reconciliation:** Partners participate in the **linear infrastructure rent and revenue**, not just the data centre construction.

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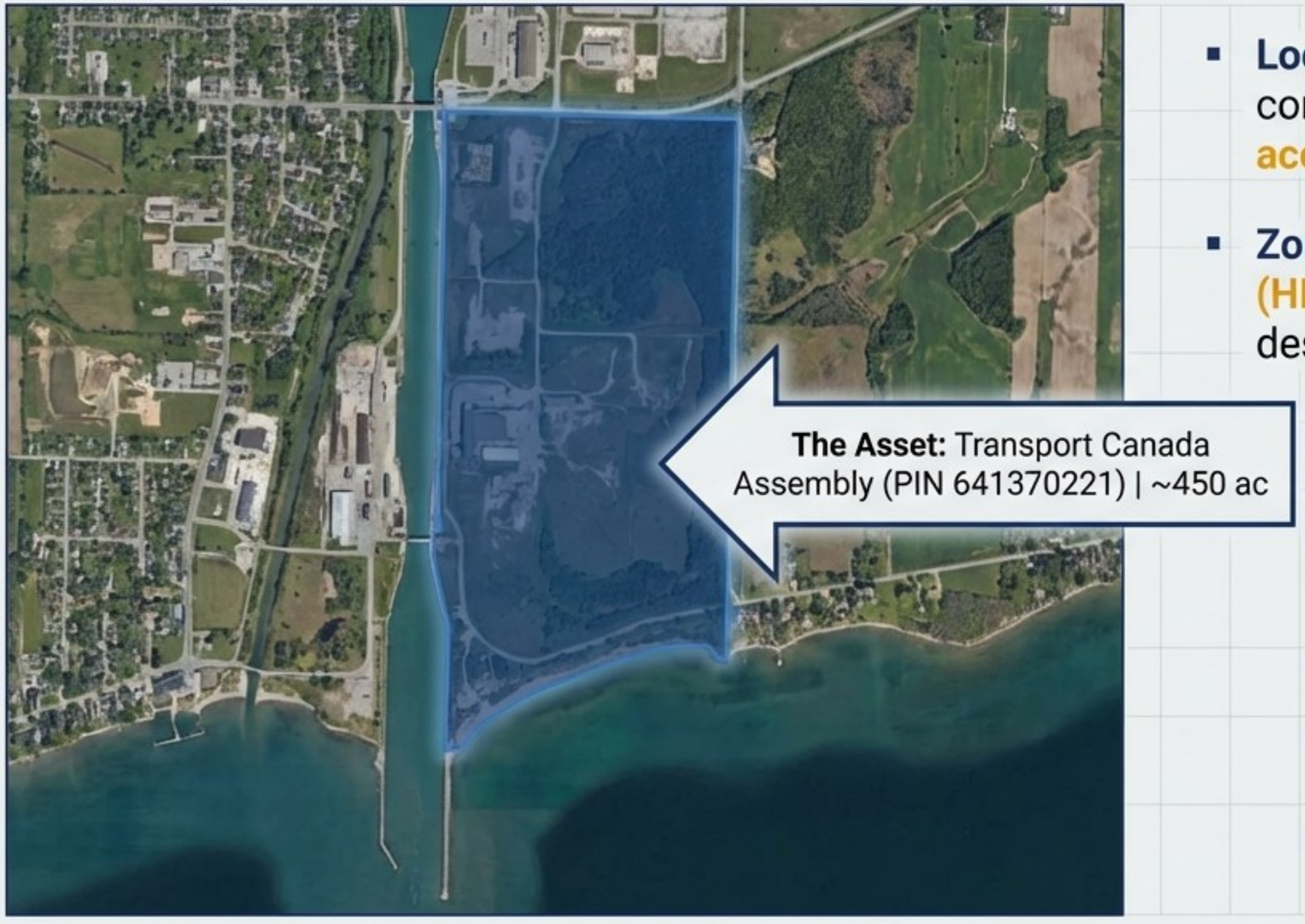
PHASE 3 – PORT COLBORNE CROWN NODE



- **Location**: Southern terminus of the corridor at Lake Erie; **direct water access for advanced cooling solutions**.
- **Zoning Status**: Mix of **Heavy Industrial (HI)** and holding zones; historically designated for **major federal works**.

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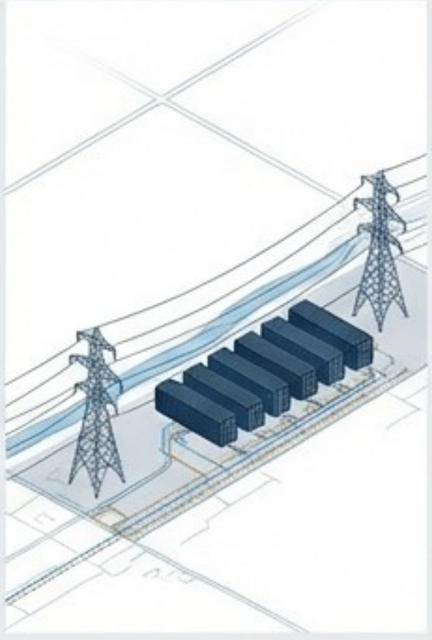
THE SOUTHERN GATEWAY STRATEGY

ALLANBURG (NORTH)

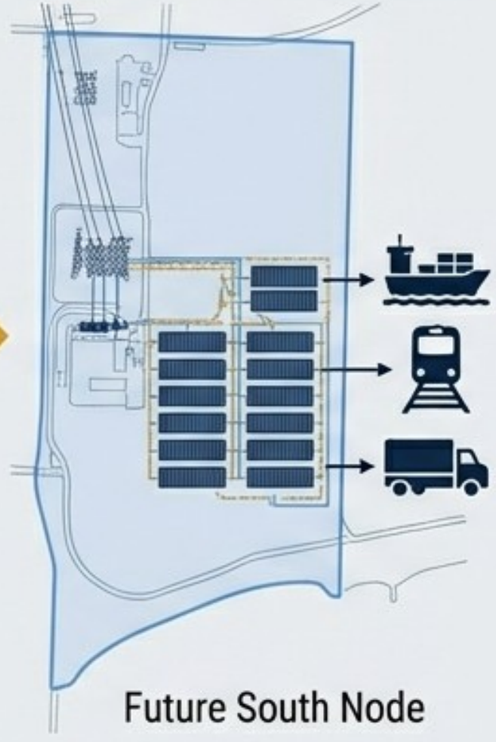


Existing North Node

Existing
North Node

SYMMETRICAL GRID ACCESS

PORT COLBORNE (SOUTH)



Future South Node

- **Power:** Short transmission hop to **Port Colborne TS**; mirrors the Allanburg setup at the southern end of the belt.
- **Scale:** Large acreage supports the massive footprint required for future **500 MW+** training clusters.
- **Logistics:** **Multimodal access** (Marine/Rail/Road) supports heavy equipment delivery for data centre construction.

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THE 'ONE CORRIDOR' ADVANTAGE

EFFICIENCY

A single federal counterparty (Transport Canada/Seaway) for the entire 3-node utility spine.

SCALABILITY

Phased growth from 100 MW (Allanburg) -> 300 MW (Welland) –> 1 GW+ (Port Colborne).

SOVEREIGNTY

Data remains on Canadian soil, powered by Canadian electrons, traveling over federally controlled land corridors.

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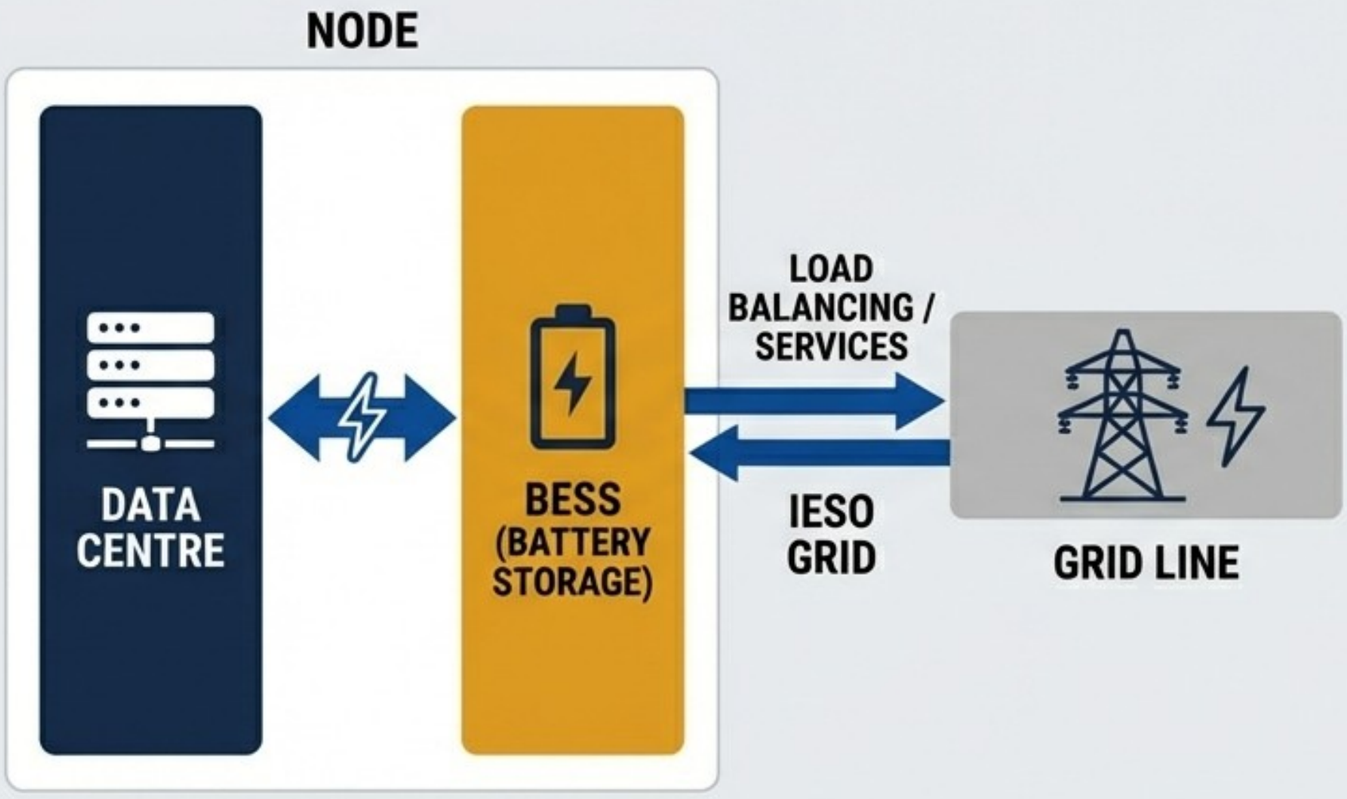
GRID & BESS INTEGRATION STRATEGY



- **LOAD BALANCING:** Each node includes BESS to smooth AI demand spikes and provide services back to the **IESO grid**.
- **CLEAN POWER:** Direct access to Niagara hydro generation + potential for on-corridor solar deployment on buffer lands.
- **RESILIENCE:** Distributed **3-node** architecture prevents single-point-of-failure risks for critical government workloads.

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KEY PARTNERS & STAKEHOLDERS

FEDERAL

ISED
(AI Strategy)

Transport Canada
(Landowner)

St. Lawrence Seaway
(Landowner)

PROVINCIAL/UTILITY

Ministry of Energy

Hydro One
(Transmission)

OPG (Land/Generation)

IESO

LOCAL & INDIGENOUS

Municipalities:
Thorold, Welland, Port Colborne

Indigenous Investment Partners

(ROW Equity)

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NEXT STEPS / THE ASK

IMMEDIATE

Confirm transmission capacity availability at Allanburg and Allanburg and Welland TS for Phase 1 & 2.

SHORT-TERM

Establish an MOU with Transport Canada/Seaway regarding utility easements and land lease options.

CALL TO ACTION

Form a "Niagara Sovereign AI Task Force" to fast-track zoning and power studies for the corridor.

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NIAGARA SOVEREIGN AI CORRIDOR

CONTACT & TEAM

Strategic Advisor / Infrastructure Lead

[Insert Contact Email/Phone]

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